

Logs, Threats and Digital clues with SPLUNK.

Turning data into detection - 5 days of threat hunting mastery.

Tool : UBUNTU / SPLUNK

Focus : Developing hands-on SOC analyst skills by detecting, investigating, and interpreting security events through log analysis, Splunk queries, and threat-hunting techniques.

Scenarios Investigated:

Day 16: Installed and configured Splunk Enterprise on Ubuntu for local and remote access.

Day 17: Zeek DNS logs to detect most queried domains and high-frequency sources.

Day 18: Investigated SSH logs to identify brute-force attempts and connection trends.

Day 19: Examined HTTP logs for server errors, suspicious user agents, and possible data exfiltration.

Day 20: Reviewed Zeek connection logs to establish network traffic baselines and detect anomalies.

Day 16: Installing & Configuring Splunk on Ubuntu

Objective:

Set up Splunk Enterprise on Ubuntu for local and remote log analysis.

Task Performed:

Downloaded and installed Splunk (v9.3.0) via wget and dpkg.

Enabled Splunk as a boot-start service and accepted the license.

Configured UFW to allow external web access on port 8000.

Edited web.conf to enable remote access from any IP.

Restarted the Splunk service and accessed via browser.

```
Ubuntu 24.04 LTS - VMware Workstation 17 Player (Non-commercial use only)
Player
Jul 15 14:51
root@reliance:/opt/splunk/bin

root@reliance:/home/reliance# wget -O splunk-9.3.0-51ccf43db5bd-linux-2.6-and64.deb "https://download.splunk.com/products/splunk/releases/9.3.0/linux/splunk-9.3.0-51ccf43db5bd-linux-2.6-and64.deb"
--2025-07-15 14:25:35-- https://download.splunk.com/products/splunk/releases/9.3.0/linux/splunk-9.3.0-51ccf43db5bd-linux-2.6-and64.deb
Resolving download.splunk.com (download.splunk.com)... 108.157.78.34, 108.157.78.4, 108.157.78.10, ...
Connecting to download.splunk.com (download.splunk.com)|108.157.78.34|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 755608332 (721M) [binary/octet-stream]
Saving to: 'splunk-9.3.0-51ccf43db5bd-linux-2.6-and64.deb'

linux-2.6-and64.deb 48%[=====] 348.82M 578KB/s eta 6 b5bd-linux-2.6-and64.de 90%[=====] 648.60M 1.07MB/s splunk-9.3
0-51ccf43d 100%[=====] 720.60M 1.45MB/s in 13m 8s

2025-07-15 14:38:45 (937 KB/s) - 'splunk-9.3.0-51ccf43db5bd-linux-2.6-and64.deb' saved [755608332/755608332]

root@reliance:/home/reliance# ls
Desktop Music Public Templates
Documents nohup.out restartfunction.sh Videos
Downloads password.txt snap
juice-shop Pictures splunk-9.3.0-51ccf43db5bd-linux-2.6-and64.deb

root@reliance:/home/reliance# sudo dpkg -i splunk-ubuntu.deb
dpkg: error: cannot access archive 'splunk-ubuntu.deb': No such file or directory
root@reliance:/home/reliance# dpkg -i splunk-9.3.0-51ccf43db5bd-linux-2.6-and64.deb
Selecting previously unselected package splunk.
(Reading database ... 266877 files and directories currently installed.)
Preparing to unpack splunk-9.3.0-51ccf43db5bd-linux-2.6-and64.deb ...
Unpacking splunk (9.3.0) ...
Setting up splunk (9.3.0) ...
complete
root@reliance:/home/reliance# cd /opt/splunk/bin/
root@reliance:/opt/splunk/bin# ls
2to3-3.7 exporttool jsnln pld_check.sh pydoc3.7 safe_restart_cluster_master.py supervisor-simulator
2to3-3.9 fullsummary_index.py locktest pip pydoc3.9 scripts tart.py
bloom genAuditKeys.py locktool pip3 python scrubber.py tocsv.py
bottle.py genRootCA.sh mongod pip3.7 python3 searchtest tsidxprobe
btool genSignedServerCert.py mongod-3.6 pip3.9 python3.7 setSplunkEnv tsidxprobe.plo
btprobe genSignedServerCert.sh mongod-4.0 prichunkpng prigraypng shc_upgrade_template.py slin tsidx_scan.py
bz2p2 genWebCert.py mongodump nongorestore primpapng pyvenv signtool untarit.py
classify ColdStorageArchiver_GCP.py idle3 node noah_self_storage_archiver.py pyvenv-3.7 splunk walklex
ColdStorageArchiver.py idle3.7 node priamtopng pripnglsch recover-metadata splunkmon wheel
ColdToFrozenExample.py idle3.9 openssl
```

```
Ubuntu 24.04 LTS - VMware Workstation 17 Player (Non-commercial use only)
Player
Jul 15 15:02
root@reliance: /opt/splunk/etc

All preliminary checks passed.
Starting splunk server daemon (splunkd)...
Generating a RSA private key
.....
writing new private key to 'privKeySecure.pem'
.....
Signature ok
subject:/CN=reliance/O=SplunkUser
Getting CA Private Key
writing RSA key
PYTHONHTTPSVERIFY is set to 0 in splunk-launch.conf disabling certificate validation for the http lib and urllib libraries shipped with the embedded Python interpreter; must be set to "1" for increased security
Done
Waiting for web server at http://127.0.0.1:8000 to be available..... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://reliance:8000

root@reliance: /opt/splunk/bin# ./splunk stop
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
...
Stopping splunk helpers...
Done.
root@reliance: /opt/splunk/bin# cd /opt/splunk/etc/
root@reliance: /opt/splunk/etc# ls
anonymizer  datettime.xml  lnttd  log-btool-debug.cfg  log-debug.cfg  master-apps  packages  searchLanguage.xml  splunk-launch.conf.default
apps  deployment-apps  instance.cfg  log-cfg  log-searchprocess.cfg  modules  packageType  shcluster  splunk-version
auth  disabled-apps  licenses  log-cndline.cfg  log-utility.cfg  myinstall  passwd  splunk-enttrial.lic  system
copyright.txt  findlogs.txt  log-btool.cfg  log-cndline-debug.cfg  manager-apps  openldap  prettyprint.xml  splunk-launch.conf  users
root@reliance: /opt/splunk/etc# nano splunk-launch.conf
root@reliance: /opt/splunk/etc#
```

```
Ubuntu 24.04 LTS - VMware Workstation 17 Player (Non-commercial use only)
Player
Jul 15 15:09
root@reliance: /opt/splunk/bin

root@reliance: /opt/splunk/etc# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.172.64  netmask 255.255.255.0  broadcast 192.168.172.255
    inet6 fe80::20c:29ff:fe11:f486  prefixlen 64  scopeid 0x20<link>
    ether 00:0c:29:11:ff:86  txqueuelen 1000  (Ethernet)
    RX packets 638874  bytes 924812843 (924.0 MB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 141616  bytes 11295106 (11.2 MB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop  txqueuelen 1000  (Local Loopback)
    RX packets 9641  bytes 3105707 (3.1 MB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 9641  bytes 3105707 (3.1 MB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

root@reliance: /opt/splunk/etc# cd /opt/splunk/bin/
root@reliance: /opt/splunk/bin# ./splunk start

Splunk> CSI: Logfiles.

Checking prerequisites...
Checking http port [0.0.0.0:8000]: open
Checking mgmt port [0.0.0.0:8089]: open
Checking appserver port [127.0.0.1:8065]: open
Checking kvstore port [0.0.0.0:8191]: open
Checking configuration... Done
Checking critical directories... Done
Checking indexes...
Validated: _audit_configtracker_dsappevent_dsclient_dsphonehome_in
ternal_introspection_metrics_metrics_rollup_telemetry_thefishbucket history main s
ummary
Done
Checking filesystem compatibility... Done
Checking conf files for problems...
Invalid key in stanza [cloud] in /opt/splunk/etc/apps/splunk_assist/def
ault/assist.conf, line 14: http_client_timeout_seconds (value: 30).
Your indexes and inputs configurations are not internally consistent. F
```

splunk>enterprise

Apps

Admin Manage

2

Messages Settings Activity Help

Fire

Search apps by name...

Search & Reporting

Splunk Secure Gateway

Upgrade Readiness App

Find more apps

Bookmarks

Dashboard

Search

My bookmarks (0) Add bookmark

Shared with my organization (0) Add boo

Splunk recommended (13)

Key Takeaways:

Splunk installation on Ubuntu is straightforward with .deb packages.

Firewall and web.conf changes are required for remote access.

Post-install configuration readies Splunk for immediate log ingestion.

Day 17: Splunk Basics – DNS Log Analysis

Objective:

Analyze DNS traffic using Zeek JSON logs in Splunk.

Task Performed:

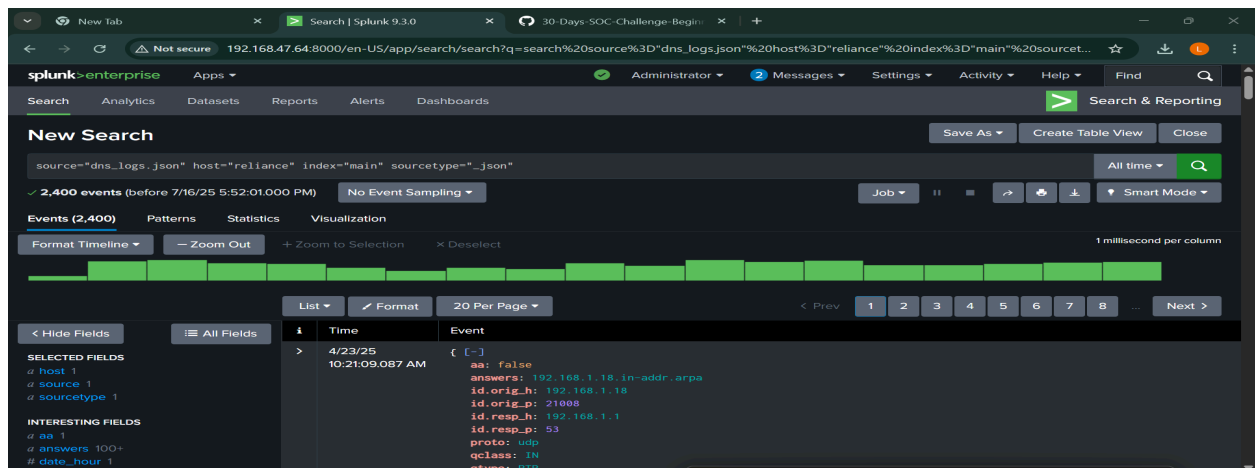
Uploaded Zeek DNS logs to Splunk (dns_lab index, json sourcetype).

Ran SPL queries to:

Identify most queried domains.

Find most active source hosts.

Display DNS query type distribution.



This screenshot shows a Splunk search interface with the query `Index=* | stats count by query | sort -count`. The results table displays the top domains by query count.

query	count
google.com	278
printer.local	278
example.com	274
yahoo.com	250
fileserver.local	248
router.local	232
internal.lan	228
backup.local	210

This screenshot shows a Splunk search interface with the query `Index=* | stats count by "id.orig_h" | sort -count`. The results table displays the top IP addresses by count.

id.orig_h	count
192.168.1.10	226
192.168.1.18	226
192.168.1.21	224
192.168.1.20	210
192.168.1.12	204
192.168.1.13	198
192.168.1.16	194
192.168.1.17	192
192.168.1.19	190
192.168.1.11	184
192.168.1.15	178

This screenshot shows a Splunk search interface with the query `Index=* | stats count by qtype`. The results table displays the counts for different query types.

qtype	count
A	626
AAAA	642
CNAME	514
PTR	618

Key Takeaways:

DNS logs are valuable for mapping normal vs suspicious traffic.

SPL's stats count by reveals high-frequency queries.

Query type analysis helps detect unusual DNS usage.

Day 18: Splunk Basics – SSH Log Analysis

Objective:

Detect failed logins, successful logins, and suspicious SSH activity.

Task Performed:

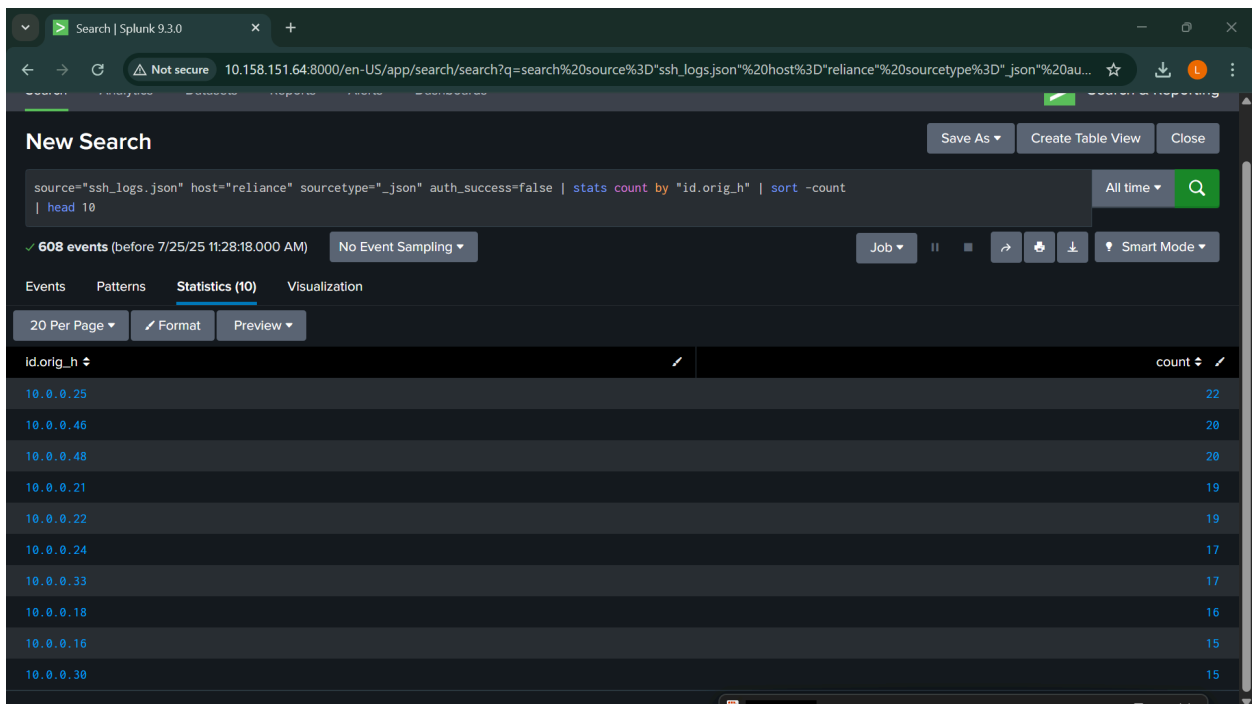
Uploaded Zeek-style SSH logs (ssh_lab index, json sourcetype).

Ran SPL queries to:

List top 10 failed login sources.

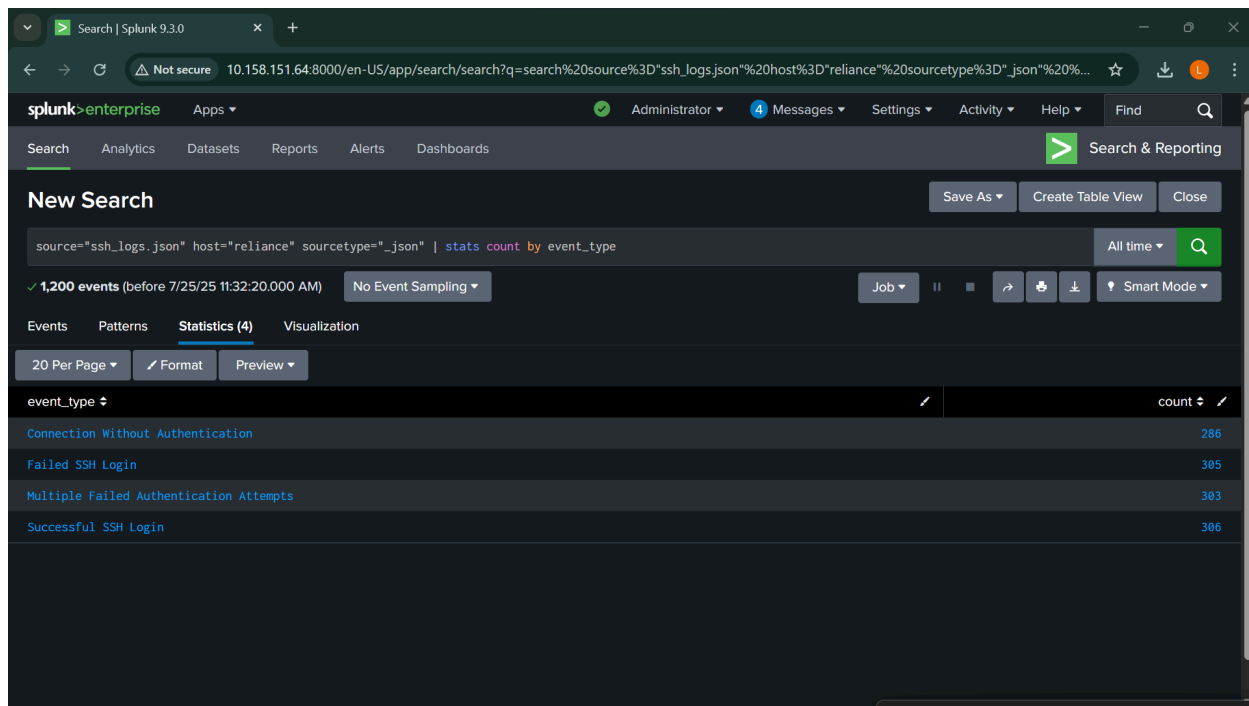
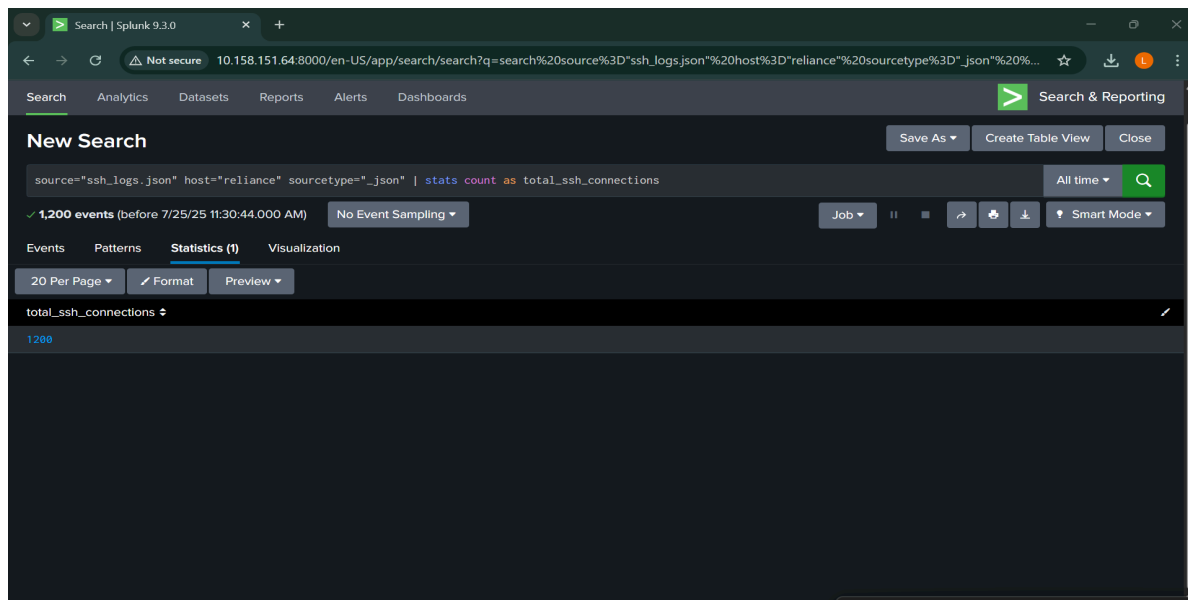
Count total SSH connections.

Summarize event types.



The screenshot shows the Splunk Search interface. The search bar contains the query: `source="ssh_logs.json" host="reliance" sourcetype="_json" auth_success=false | stats count by "id.orig_h" | sort -count | head 10`. The results show 608 events. The table displays the top 10 failed login sources by IP address (id.orig_h) and their count.

id.orig_h	count
10.0.0.25	22
10.0.0.46	20
10.0.0.48	20
10.0.0.21	19
10.0.0.22	19
10.0.0.24	17
10.0.0.33	17
10.0.0.18	16
10.0.0.16	15
10.0.0.30	15



Key Takeaways:

Brute force attempts often appear as clustered failed logins.

Total connection counts give baseline activity levels.

Event type breakdown helps prioritize investigation.

Day 19: Splunk Basics – HTTP Log Analysis

Objective:

Investigate HTTP activity to find errors, suspicious behavior, and large transfers.

Task Performed:

Uploaded Zeek-style HTTP logs (http_lab index, json sourcetype).

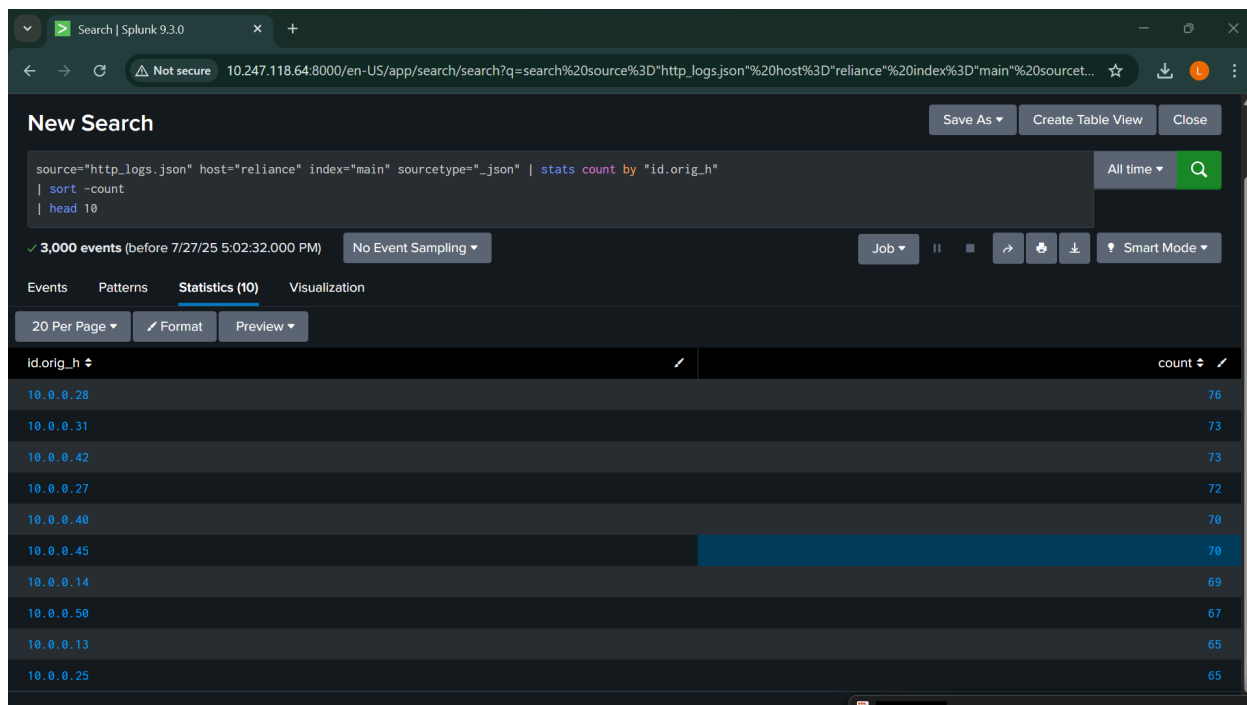
Ran SPL queries to:

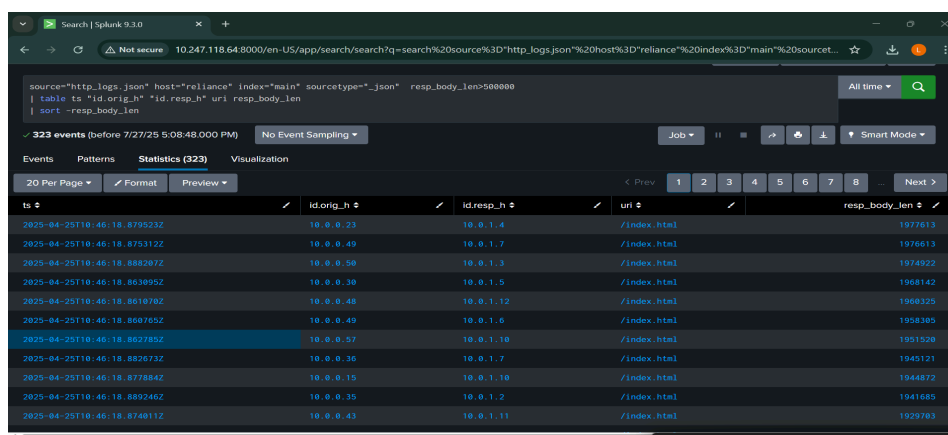
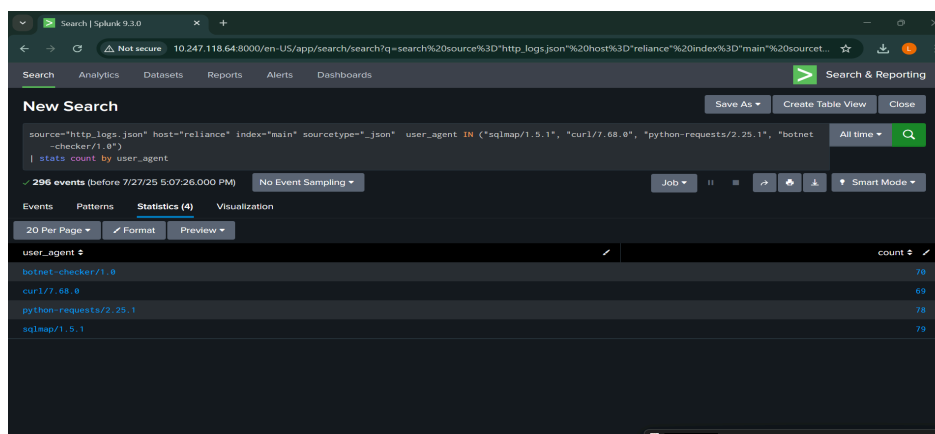
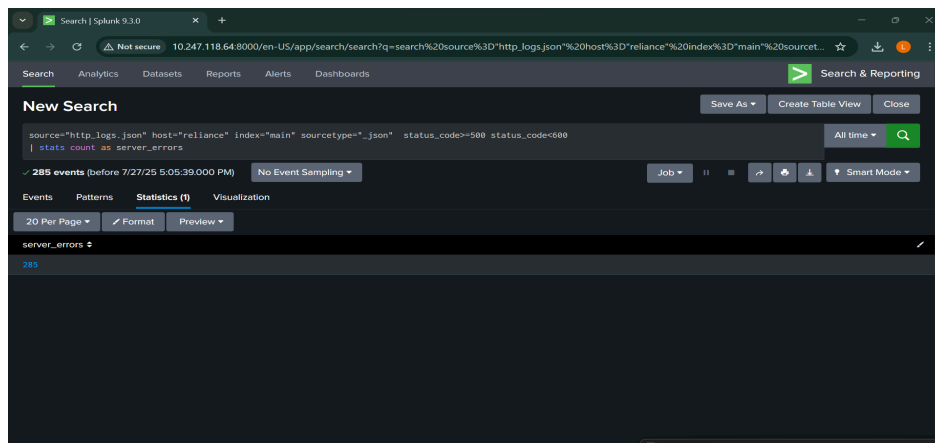
Find top HTTP request sources.

Count server errors (5xx).

Identify suspicious user agents.

Detect large file transfers (>500KB).





Key Takeaways:

HTTP logs reveal both operational errors and attack patterns.

Known malicious user agents can quickly flag threats.

Large transfers may indicate data exfiltration.

Day 20: Splunk Basics – Zeek Connection Log Analysis

Objective:

Analyze Zeek connection logs to identify traffic patterns and high-usage endpoints.

Task Performed:

Uploaded Zeek connection logs (conn_lab index, json sourcetype).

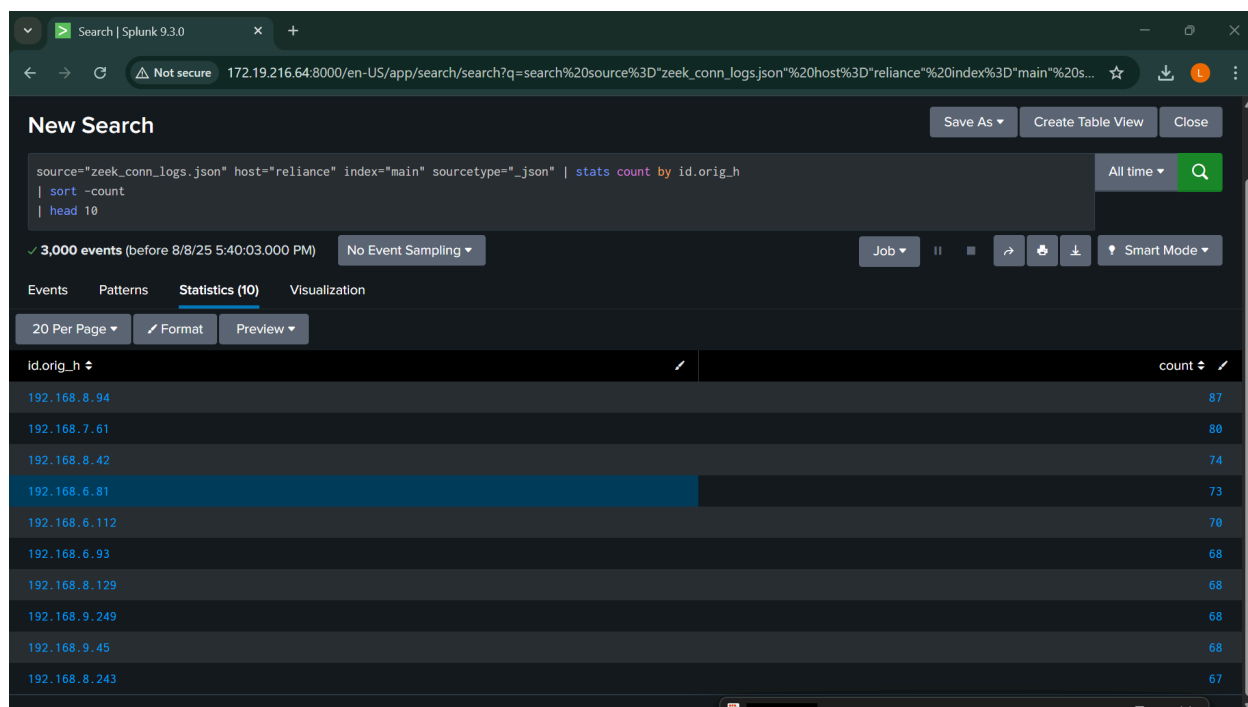
Ran SPL queries to:

Identify top client IPs.

List most common services.

Find long-duration sessions (>1s).

Detect most accessed internal servers.



The screenshot shows the Splunk Search interface. The search bar contains the query: `source="zeek_conn_logs.json" host="reliance" index="main" sourcetype="_json" | stats count by id.orig_h | sort -count | head 10`. The results show 3,000 events. The table below lists the top client IPs and their counts.

id.orig_h	count
192.168.8.94	87
192.168.7.61	80
192.168.8.42	74
192.168.6.81	73
192.168.6.112	70
192.168.6.93	68
192.168.8.129	68
192.168.9.249	68
192.168.9.45	68
192.168.8.243	67

New Search

source="zeek_conn_logs.json" host="reliance" index="main" sourcetype="json" | stats count by service
| sort -count

3,000 events (before 8/8/25 5:41:31.000 PM) No Event Sampling

Events Patterns Statistics (10) Visualization

20 Per Page Format Preview

service	count
dns	395
dhcp	378
snmp	369
ntp	348
ldap	271
ssh	263
smtp	254
ftp	251
http	247
imap	232

New Search

source="zeek_conn_logs.json" host="reliance" index="main" sourcetype="json" | stats count by service
| sort -count

3,000 events (before 8/8/25 5:41:31.000 PM) No Event Sampling

Events Patterns Statistics (10) Visualization

20 Per Page Format Preview

time	source	dest	service	count
10.0.0.1	10.0.0.2	10.0.0.3	dns	1
10.0.0.1	10.0.0.2	10.0.0.3	dhcp	1
10.0.0.1	10.0.0.2	10.0.0.3	snmp	1
10.0.0.1	10.0.0.2	10.0.0.3	ntp	1
10.0.0.1	10.0.0.2	10.0.0.3	ldap	1
10.0.0.1	10.0.0.2	10.0.0.3	ssh	1
10.0.0.1	10.0.0.2	10.0.0.3	smtp	1
10.0.0.1	10.0.0.2	10.0.0.3	ftp	1
10.0.0.1	10.0.0.2	10.0.0.3	http	1
10.0.0.1	10.0.0.2	10.0.0.3	imap	1

New Search

source="zeek_conn_logs.json" host="reliance" index="main" sourcetype="json" | stats count by "id.resp_h"
| sort -count
| head 10

3,000 events (before 8/8/25 5:44:29.000 PM) No Event Sampling

Events Patterns Statistics (10) Visualization

20 Per Page Format Preview

id.resp_h	count
10.0.0.1	245
10.0.0.2	273
10.0.0.5	268
10.0.0.10	257
10.0.0.4	254
10.0.0.7	252
10.0.0.3	258
10.0.0.5	258
10.0.0.8	245
10.0.0.12	235

Key Takeaways:

Network activity is often concentrated among a few clients and services.

Long-duration SSH/HTTP sessions may require deeper review.

Zeek connection logs are a rich source for baseline network behavior.

SUMMARY

During Days 16–20, the focus was on enhancing log ingestion, enrichment, and analysis skills using Splunk. I set up Splunk Enterprise on Ubuntu, configured both local and remote access, and verified ingestion of Apache Access logs, Windows Security Event logs, Suricata alerts, and Zeek DNS/HTTP/Connection logs with correct source types for accurate field extraction.

Investigations included reviewing Zeek DNS logs to identify top queried domains and detect clients with abnormal request volumes, suggesting beaconing or misconfigurations. SSH log analysis revealed repeated failed login attempts indicative of brute-force attacks, which I correlated with successful logins for compromise checks. HTTP logs were analyzed for spikes in 4xx/5xx errors, unusual traffic patterns, and suspicious user-agent strings.

Connection logs were examined for top network talkers, long-lived sessions, and unusual ports, flagging possible exfiltration or C2 activity. Establishing baseline traffic patterns also improved anomaly detection.

These activities sharpened my SPL querying skills, strengthened pivoting between log types, and reinforced the importance of correlating multiple data sources to uncover the full scope of potential incidents.